Form of Explanation

Explainability vs. Explicability under Multitasking Load

Does the form of an automation aid's reasoning explanation — holding action-relevant information constant — change subjective transparency and *whether the operator's mental model actually updates under multitasking load?*

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Problem & Relevance

Two systems can be equally transparent — yet only one updates the operator's mental model.

Problem

When an AI system explains itself, three things come together:

- **Transparency** gives the operator insight into the system's inner workings
- **Explainability** is the system's act of producing an explanation of what it did and how
- **Explicability** is whether that explanation is presented in a way that it is actually understood and processed by the operator.

We want to focus on the effects of different **forms of an explanation** while holding the action-relevant information constant.

Relation to the lecture content

The project is anchored in two lectures:

- **Transparency**: The question of what level of information the aid should communicate, Three-Gap-Model
- **Explainability & Explicability**: Criteria for good explanations, among others selectivity, contrastiveness and actionability.

Why it is relevant

This topic matters wherever a human supervises an automated system — **aviation, medical monitoring, autonomous driving aids.**

If an explanation is long and reads like a status log, the operator may feel informed without being able to predict the system's next action. This may lead to the human supervisor missing automation errors with far-reaching consequences.

Research Gap

Transparency studies (Mercado 2016, Stowers 2017) vary the *level* of information but do not distinguish information **content** and **presentation**. Our design holds action-relevant content and varies **only the form**, which isolates the presentation aspect.

The XAI literature shows that explanations can raise agreement and the feeling of being informed without improving the user's ability to catch AI mistakes (Bansal 2021), and that subjective understanding often rises without objective understanding (Wang & Yin 2021) — but these results come from single-task settings. We test the same dissociation under **multitasking load**.

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Concept — three forms of explanation

All three forms communicate the aid's reasoning. The comparison is therefore between forms of explanation, not between explanation and no explanation.

F1

Verbose / always-on

Routine status update, every cycle.

Example panel text

```
"Scale-1: was 47.3 > upper bound 45.0 for
1.0 s - reset to 32.0. Scale-2: 31.5, in
range. Scale-3: 29.7, in range. Scale-4:
30.2, in range. Lights: OK.“

“Scale-1: was 47.0 > upper bound 45.0, auto-
aid not engaged. Scale-2: 31.5, in range.
Scale-3: 29.7, in range. Scale-4: 30.2, in
range. Lights: OK.”
```

Embeds every fact in routine update — fires on every automation cycle.

F2

Selective + contrastive

Panel only on near-miss / miss.

Example panel text

```
"Reset scale-1 - would have failed in ~2.5
s.“

"Skipped scale-2 – auto-aid did not act."
```

Same necessary action-relevant facts as F1, reframed selectively and contrastively.

F3

Selective + contrastive + actionable

F2 plus an explicit operator cue on misses.

Example panel text

"Reset scale-1 – would have failed in ~2.5 s.“

"Skipped scale-2 – auto-aid did not act. **Check scale-2 manually.**"

F2 plus an explicit operator cue.

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Research Question & Hypotheses

H1 + H2 jointly instantiate the explainability ≠ explicability crossover; H3 and H4 are supporting.

Research question

Does the form of an automation aid's reasoning explanation — holding action-relevant information constant — change subjective transparency and whether the operator's mental model actually updates under multitasking load?

Hypotheses · H1 and H2 are the main hypothesis

H1 Subjective transparency and subjective trust favour verbose explanations

Participants will rate the verbose always-on explanation (F1) as more transparent and more trustworthy than the selective explanations (F2/F3) , even when action-relevant information is held constant.

H2 Form modulates explicability

Participants exposed to selective and contrastive explanations (F2/F3) will achieve higher post-block mental-model accuracy than participants exposed to a verbose always-on explanation (F1), even when action-relevant information is held constant.

H3 Selectivity directs attention to anomalies and improves subjective trust

A: Participants will detect a higher proportion of automation-missed events under selective explanations (F2/F3) than under a verbose always-on explanation (F1).

B: Participants will less likely overwrite the aid even when it was right under (F2/F3) than under (F1).

H4 Saved attention is redeployed

Participants will achieve higher performance on the non-automated tasks (communications and resource management) under selective explanations (F2/F3) than under a verbose always-on explanation (F1).

Method

Within-subjects on the OpenMATB platform · ~20 min per participant.

Study design and implementation

Platform. OpenMATB — three concurrent supervisory tasks: system monitoring, communications, and resource management.

Automation aid. Runs **only** on system monitoring at ~78% reliability; the other two tasks compete for attention and are not aided. Every automation miss is communicated, so that the operator can handle the event manually if the information is perceived and processed.

Within-subjects design. Each participant runs all three forms (F1/F2/F3), each on three different gauge sets (A,B,C). The gauge sets contain different subsets of gauges and lights that need the operator's attention, so the post-block memory probe cannot be answered by recall from another block. Order of F1/F2/F3 is counterbalanced across participants to mitigate learning effects.

Session. ≈ 20 min per participant: 3 min practice → three 5 min experimental blocks → debriefing.

Practice | **Block 1** | **Block 2** | **Block 3** | **Debrief**

3 min → 5 min → 5 min → 5 min → questionnaires

What we measure

- Three questionnaires after each block
- Platform log
- One final questionnaire

Content matching.

F1's verbose stream contains every fact F2/F3 reports. All information needed by the operator to catch automation failure is equally present in F1, F2, and F3.

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Measures & Considerations

Each measure maps onto a specific hypothesis.

Mental-model probe

H1, H2

Questionnaire · after each block

Five short slider questions on what the participant believes the aid actually did (e.g., "how many gauges did it act on?", "how often did it skip a gauge that needed attention?"). Compared against the scenario ground truth — an objective explicability score.

Detection of automation misses

H3

Platform log

Whenever the aid skips an event, the participant should catch it manually. We record hit/miss and reaction time.

Detection of automation aid overwrites

H3

Platform log

Whenever the aid handles an event, the participant should not intervene. We this record interventions the user persons although the automation aid catches the event.

Subjective transparency

H1

Questionnaire · after each block

Three ratings: "the system told me what it was doing", "I understood why the system acted or did not act", "the amount of information felt about right". Captures the feeling of being informed — paired with the probe to test the H1 dissociation.

Subjective trust

H1

Questionnaire · after each block

1.Several rating like “The aid is dependable.”, “The aid is reliable.”, “I am confident in the aid.”, … to capture subjective trust.

Non-automated task performance

H4

Platform log

Communications and resource management are not aided — so they reveal whether reading less in the panel actually frees attention elsewhere. We are measuring performance in these tasks.

Considerations &Limits

- F1 → F2 changes three factors simultaneously (selectivity + contrastiveness + framing). Treated as a package-level test of "form", not as three isolated effects; F3 – F2 isolates the actionable cue.
- Load is held at a single moderate level; generalisation across load levels cannot be done since the load level is held constant.
- The number of automation aid misses is small (3) in the 5 min block. Reliable statistic on such small number is difficult.
- The number of participants might be to low to see any effect with significance (Number of participants in the literature is at least 50-100).

References

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